

Antoine Vuignier

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Final-year PhD candidate in theoretical physics at EPFL. My work focuses on turning abstract questions into well-defined mathematical problems and producing quantitative answers, using a combination of analytical reasoning and computational methods. Seeking a research-oriented position in machine learning, applied science, or scientific computing.

Education

- 2022 – ... ♦ **PhD, EPFL - Fields and Strings Laboratory**
Supervised by João Penedones. Research on the emergence of spacetime in quantum gravity and non perturbative approaches to quantum field theory. Combined pen-and-paper calculations with computational workflows in Mathematica and Python for symbolic algebra, numerical linear algebra, and large-scale convex optimization (semidefinite programming) on external clusters.
Thesis: *What is spacetime made of?*
- 2017 – 2022 ♦ **Bachelor and Master of Physics, EPFL**

Skills

- General ♦ Problem solving, critical thinking, public speaking, teaching, project supervision.
- Languages ♦ Native in French, fluent in English, B2 in German.
- Coding ♦ Python (NumPy, SciPy, PyTorch, JAX, matplotlib), C++, Mathematica, $\LaTeX$. Experience with numerical optimization, symbolic computation and cluster based workflows.

Research Output

- 2022-2025 ♦ 5 publications in high energy theoretical physics (JHEP), see INSPIRE profile <https://inspirehep.net/authors/2612961>.
- 2024-2025 ♦ Multiple invited talks at international workshops and universities

Training and Projects

- 2025-2026 ♦ **ML Projects** - Built a neural network approach (PyTorch) to solve matrix models. Also built a progression of models from scratch in PyTorch to learn deep learning fundamentals: CNN for digit recognition (JAX/Flax + NumPy-only implementation), character-level language model with knowledge distillation (PyTorch), and a GPT-2-style Transformer with multi-head causal attention (PyTorch). Code on GitHub.
- 2025 ♦ **Artificial Intelligence for High Energy Physics** - Hands-on training with modern ML frameworks (JAX). Studied neural network architectures, optimization methods, and their applications to physics problems.
- 2022-2025 ♦ Attended 10+ international schools and workshops in theoretical physics and mathematics.

Teaching and scientific activities

- 2019-2025 ♦ **Teaching Assistant, EPFL** - 10+ courses on physics and mathematics at the bachelor and master level. Taught at summer schools for high school students. Supervised two master students on their thesis.
- 2022-2025 ♦ Organized weekly research seminars. Gave outreach talks to high school and first year university students.